How to Develop Key Intelligence Questions and Problem Framing

A Quanta Analytica training document for junior analysts and general audiences

DOMAIN	Analytic Tradecraft Training	PRODUCER	Quanta Analytica | MNS Consulting
FRAMEWORK	QAP-aligned problem framing workflow	DATE	April 23, 2026
AUDIENCE	Junior analysts and general learners	CLASSIFICATION	Open Source
FOCUS	KIQs, source collection, workflow, and LLM integration	FORMAT	DOCX training guide

Analytic Scope Boundary: This training document explains how to turn broad topics into decision-relevant intelligence problems, draft strong Key Intelligence Questions, collect credible online sources, organize an evidence workflow, and use LLM platforms inside a governed analytic process. It is educational in purpose and draws on open web references plus Quanta Analytica internal methodology materials.

Produced under the Quanta Analytica Process™. All claims are traced to cited references or identified as internal methodology guidance.

01 // PURPOSE

Start with the decision, not the topic

1. **A topic is not yet an intelligence problem.** A junior analyst often starts with a subject area such as public health, cyber risk, or market instability. The analytic task begins only when the topic is tied to a decision, a specific uncertainty, a scope boundary, and a time horizon. [1] [2]
2. **Problem framing converts ambiguity into a structured analytic container.** Within the Quanta Analytica Process, problem framing comes before generative output and includes Key Intelligence Questions, assumption testing, and domain mapping so that analysis is driven by structure rather than drift. [8] [9]
3. **Strong KIQs reduce uncertainty that matters to a real decision.** A useful KIQ does more than describe the situation. It helps explain drivers, test assumptions, reveal decision thresholds, and establish indicators that can be monitored over time. [2] [4]
4. **LLMs should sharpen the workflow, not replace evidence handling.** Good analytic use of LLMs means constraining the model, separating facts from inferences, and keeping verification, source validation, and final judgment under human control. [3] [5] [7]

02 // FRAMING

What problem framing actually does

Topic A broad subject area such as fragile governance, hospital strain, supply chain disruption, or insider risk.	Issue A narrowed concern: conditions are worsening, outcomes are becoming unstable, or a decision window is approaching.
Problem A one-sentence statement that names the uncertainty, the decision impact, the place or domain, and the time horizon.	Decision The concrete choice the audience may need to make: allocate resources, change posture, prioritize monitoring, or select an intervention.
Scope The boundary of what is in and out: geography, actor set, operating environment, and excluded streams.	Time Horizon The period the assessment covers and the forward-looking interval that matters to the reader.

Topic to analytic problem

Weak starting point **Research health risks in Country X** This is still a subject heading. It does not tell you what uncertainty matters, what evidence would be useful, or what decision the answer should support.	Better framing **Assess whether disease outbreaks, utility instability, and health-system constraints are likely to raise operational risk over the next 90 days** This version introduces causal drivers, a scope, a decision-relevant outcome, and a time boundary. That is the start of an intelligence problem. [8] [10]

Reusable formula

```
How is [system, actor, or change] affecting [decision-relevant outcome] in [place or
domain] over [time horizon], and what would indicate meaningful deterioration or
improvement?
```

03 // KIQS

How to build strong Key Intelligence Questions

QUESTION TYPE	WEAK VERSION	STRONGER VERSION	WHY THE STRONGER VERSION WORKS
Maritime security	What is happening in the Red Sea? Too broad and largely descriptive.	Which actors have the intent and operational capability to sustain disruption of commercial shipping in the Red Sea over the next 90 days?	It names the actor problem, the type of risk, and the time horizon.
Corporate cyber risk	How bad is cyber risk? Not measurable and not tied to action.	Which business units face the highest operational exposure from third-party software dependency, and what conditions would turn a vendor outage into a continuity event?	It ties the question to exposure, impact, and decision thresholds.
Healthcare operations	Why are hospital wait times rising? Still too general.	Are rising wait times being driven primarily by workforce shortages, demand spikes, or process bottlenecks, and which driver is most actionable within the next quarter?	It creates competing explanations and forces prioritization.

KIQ design test

Keep the question if	Rewrite the question if
• Answering it reduces uncertainty tied to a real decision. • Evidence can plausibly be collected to answer it. • The scope is bounded by place, domain, and time. • The answer can map to a judgment, indicator, or threshold.	• It can be answered by a generic encyclopedia paragraph. • It has no time horizon. • It would lead to endless research with no clear stopping point. • It describes a topic but does not reduce decision uncertainty.

04 // WORKFLOW

The five-step problem framing method

01 Define the decision context

Identify the audience, the decision they may take, the cost of being wrong, and the forward horizon that matters.

02 Write a one-sentence problem statement

Turn the topic into a precise framing sentence that names the uncertainty, the decision effect, the scope, and the time horizon.

03 Map the system before explaining it

List actors, resources, dependencies, chokepoints, constraints, and baseline conditions so you do not jump into premature explanation. [10]

04 Draft 3 to 6 KIQs

Use a small set that covers the current situation, drivers, trajectory, decisions, and indicators. More questions do not equal better tradecraft.

05 Build the assumptions register

Write down the assumptions already embedded in the framing and track why they matter, what would falsify them, and what happens if they fail. [4]

05 // SOURCES

How to collect credible online sources and organize the evidence

Tier 1: Primary or authoritative

- Government ministries, regulators, court records, official statistics portals
- Multilateral bodies such as WHO, World Bank, IMF, OECD, and UN agencies
- Corporate primary filings and audited disclosures

Tier 2: High-quality secondary

- Major wire services and reputable investigative desks
- Peer-reviewed journals and transparent research institutes
- Technical NGOs with clear methodology

Tier 3: Context or lead-only

- Trade press, expert blogs, and social platforms
- Useful for leads, narrative analysis, and context
- Not suitable as sole support for load-bearing judgments without independent verification

Collection workflow

STEP	WHAT TO DO	WHY IT MATTERS
1	Convert each KIQ into specific information requirements.	Collection becomes purposeful instead of random browsing.
2	Capture source metadata immediately: title, URL, author, date, retrieval date, and source type.	Prevents citation loss and supports auditability. [1] [8]
3	Extract atomic claims rather than saving vague article summaries.	Atomic claims are easier to compare, corroborate, and challenge.
4	Test independence before counting corroboration.	Several articles that cite the same upstream report still count as one stream. [8] [9]
5	Record counterevidence and unresolved conflicts.	Strong analysis stress-tests hypotheses instead of collecting confirming evidence only. [2] [4]

Recommended project structure

Folders	Spreadsheet tabs
• 01_scope • 02_source_register • 03_notes_and_claims • 04_analysis • 05_draft_product • 06_archived_sources	• KIQ tracker • Source register • Claim log • Assumptions register • Indicators and warnings

A clean workflow protects analytic memory. A pile of browser tabs does not.

06 // LLM INTEGRATION

How to leverage LLM platforms without weakening the analysis

Good uses	High-risk misuses	Human responsibilities
• Draft problem statements and candidate KIQs • Generate search strings and information requirements • Extract atomic claims from source text you provide • Compare sources for agreement, contradiction, and gaps • Convert structured notes into cleaner prose	• Letting the model invent sources or speak with unsupported certainty • Using the model as a substitute for collection and verification • Allowing facts, assumptions, and inferences to blur together • Accepting the first generated answer as the final judgment • Using a generic prompt with no role, scope, or output constraints	• Source validation and corroboration • Confidence assignment • Final tradecraft judgment • Handling uncertainty and caveats • Ensuring the output remains decision-relevant

Ready prompt pattern

```
Role: Act as a structured analytic assistant.

Task: Convert the topic below into one problem statement, five candidate KIQs, a
draft assumptions register, and a list of information requirements.

Context: [Insert topic, audience, scope, geography, and time horizon]

Rules:
- Separate facts, assumptions, and inferences
- Do not invent sources
- Keep KIQs decision-relevant
- Bound all outputs by geography and time
- Return the result in a clean table
```

07 // EXAMPLE

Worked example: from topic to analysis workflow

Topic

Disease outbreaks and infrastructure strain in a fragile state

At the topic stage, the analyst still lacks a decision frame.

Better problem statement: How are disease outbreaks, utility instability, and public health capacity constraints interacting to raise operational and humanitarian risk over the next 90 days, and what indicators would signal localized stress becoming a broader system emergency?

Sample KIQ set

- Which outbreak trends are confirmed versus rumored?
- Is deterioration being driven primarily by disease spread, infrastructure failure, or access constraints?
- Which regions face the highest probability of service disruption within 30 to 90 days?
- What indicators would show movement from manageable pressure to systemic breakdown?
- Which interventions are realistic under current resource and governance constraints?

08 // PRACTICE

Exercises for a junior analyst

Exercise 1: Rewrite the topic

Take one vague topic from your own domain and rewrite it into a one-sentence analytic problem using a decision, a scope, and a time horizon.

Exercise 2: Build five KIQs

Draft one situation KIQ, one driver KIQ, one trajectory KIQ, one decision KIQ, and one warning KIQ.

Exercise 3: Create a source plan

List the primary, secondary, and lead-only sources you would collect first. Note where independence could be weak.

Exercise 4: Stress-test with an LLM

Use a constrained prompt to challenge your assumptions and identify missing information requirements. Do not ask the model for final truth claims.

09 // REFERENCES

Reference list

[1] Office of the Director of National Intelligence. (2015). Intelligence Community Directive 203: Analytic Standards. dni.gov/files/documents/ICD/ICD-203.pdf

[2] Heuer, R. J. (1999). Psychology of Intelligence Analysis. Center for the Study of Intelligence. cia.gov/resources/csi/static/Pyschology-of-Intelligence-Analysis.pdf

[3] National Institute of Standards and Technology. (2024). Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile (NIST AI 600-1). nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf

[4] Central Intelligence Agency. (2009). A Tradecraft Primer: Structured Analytic Techniques for Improving Intelligence Analysis. cia.gov/resources/csi/static/Tradecraft-Primer-apr09.pdf

[5] OpenAI. (n.d.). Best practices for prompt engineering with the OpenAI API. help.openai.com/en/articles/6654000-best-practices-for-prompt-engineering-with-the-openai-api

[6] OpenAI. (n.d.). Prompt engineering best practices for ChatGPT. help.openai.com/en/articles/10032626-prompt-engineering-best-practices-for-chatgpt

[7] OpenAI. (n.d.). Prompt guidance for GPT-5.4. developers.openai.com/api/docs/guides/prompt-guidance

[8] MNS Consulting. (2026). The Quanta Analytica Process™: A framework-governed intelligence operating system. *Internal project reference.*

[9] MNS Consulting. (2026). QAP Standard Operating Procedures for Source Validation and Best Practices for Internet Sources. *Internal project reference.*

[10] MNS Consulting. (2026). The Quanta Analytica Five-Layer Strategic Cognition Model. *Internal project reference.*